



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

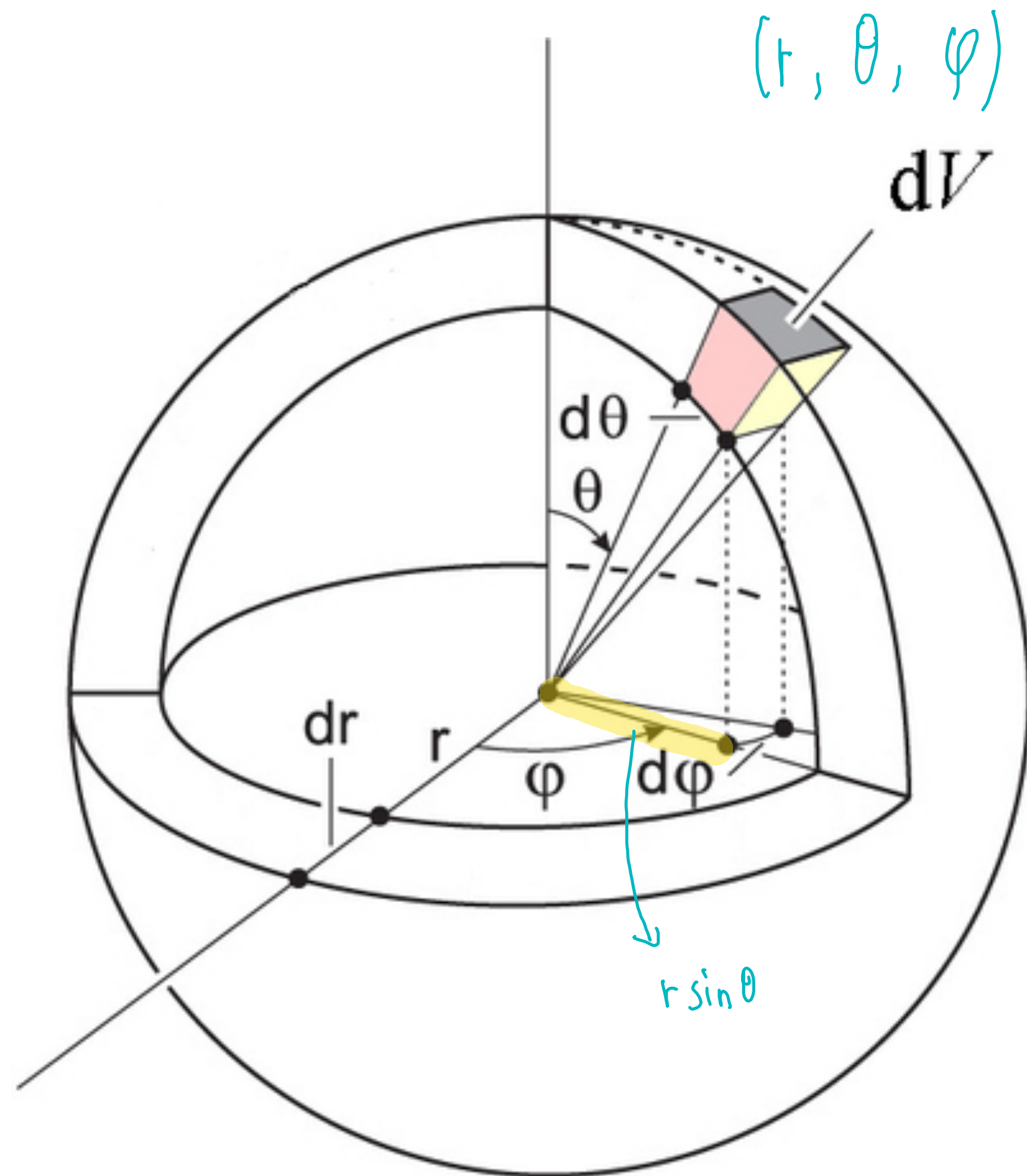
PHY1001 Mechanics

Jingqiang Ye
School of Science and Engineering (SSE)
March 6th, 2025

Midterm

- Date: **March 29th, 2025** (Saturday)
- Time: **1:30 PM – 4:30 PM** (This is the final time!)
- Coverage: All topics from the first half of the semester (from Measurements to Rotation, excluding Rolling, Torque, and Angular Momentum).
- If you have a time of conflict, please email Prof. Xiaolu Zhuo (zhuoxiaolu@cuhk.edu.cn) with Prof. Han Ling (linghan@cuhk.edu.cn) and Prof. Jingqiang Ye (yejingqiang@cuhk.edu.cn) **cced before midnight March 22nd, 2025.**
- You must provide an official document to clearly state the reason, date, time and location of this conflict.

Spherical Coordinates



Differential element

$$dV = r^2 \sin \theta dr d\theta d\phi$$

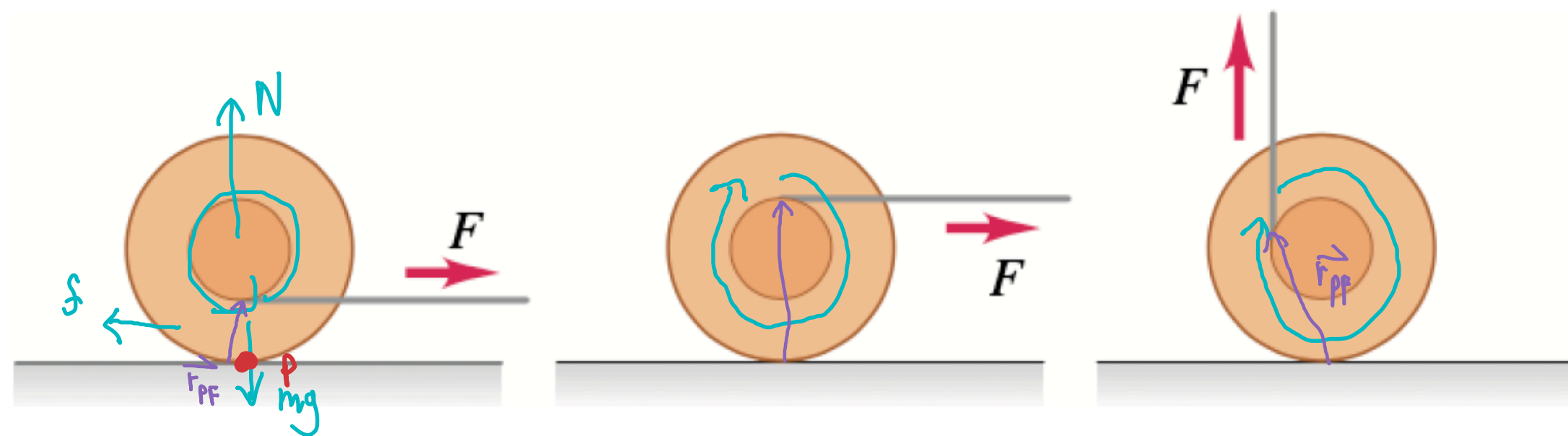
$$\begin{cases} z = r \cdot \cos \theta \\ x = r \cdot \sin \theta \cdot \cos \phi \\ y = r \sin \theta \cdot \sin \phi \end{cases}$$

Newton's 2nd Law (Rotation Version)

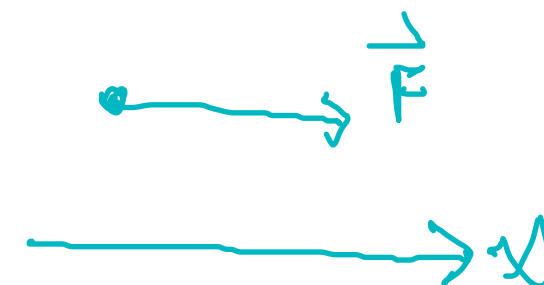
- $\tau_p = I_p \alpha$, where
 - τ_p : torque w.r.t to axis p
 - I_p : MOI w.r.t. to axis p
 - α : angular acceleration
- Axis p must be the same for torque and MOI!
- Angular acceleration α is the same (independent of axis)
- You have the freedom to choose axis!

Yo-Yo

A yo-yo is placed on the surface at rest, and then starts to roll without slipping under force F in each case. Find the direction of rotation in each case. (Hint: find the magic axis.)

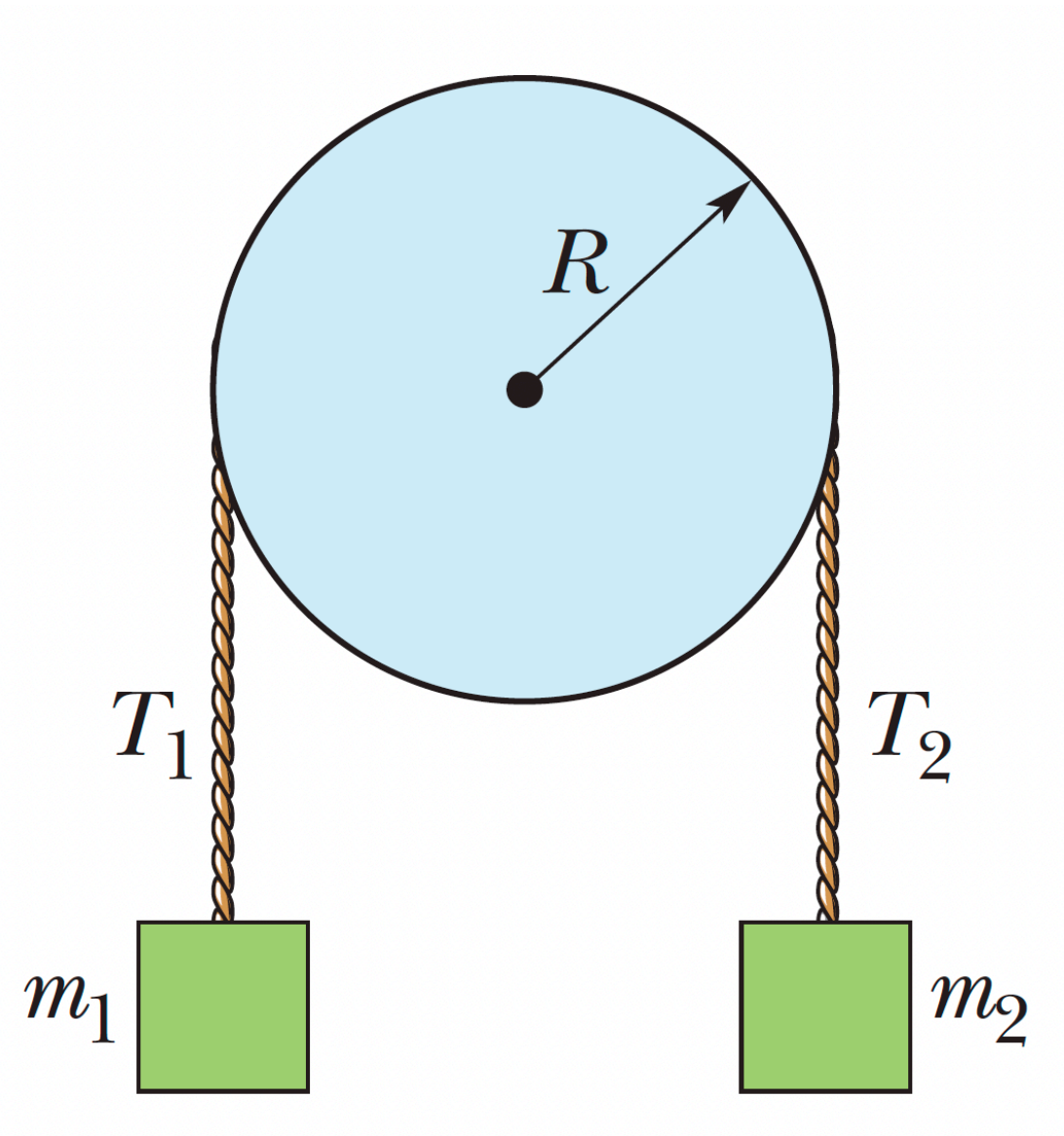


$\vec{\tau}_P$ into page
 $\vec{a}$ into page
 rotation CW



Exercise

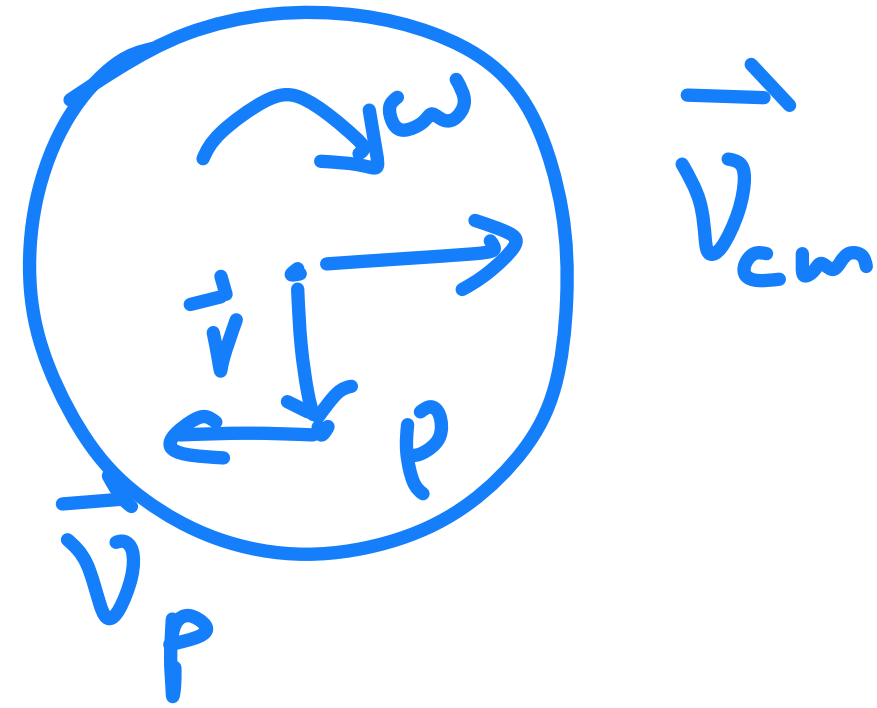
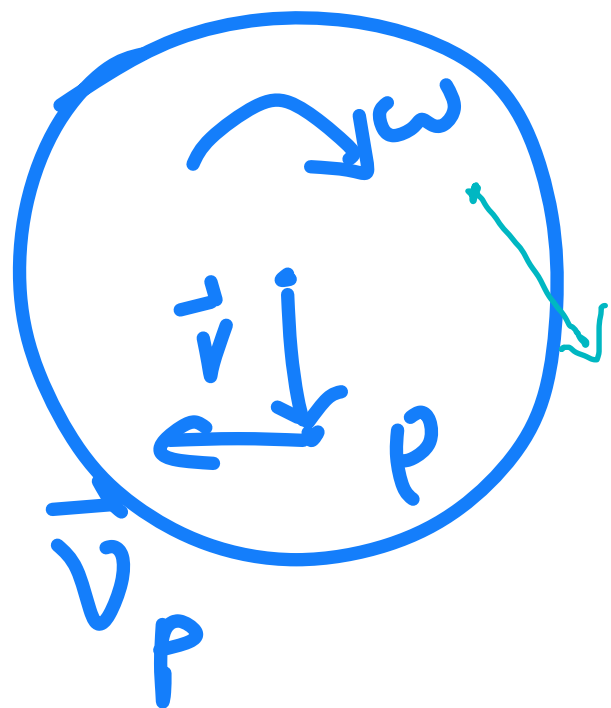
Block 1 has mass m_1 , block 2 has mass m_2 , and the pulley, which is mounted on a horizontal axle with negligible friction, has mass m_3 and radius R . When released from rest, block 2 falls a height of h without the cord slipping on the pulley. Find the velocity of m_2 .



Rotation of Rigid Body

- Velocity of point P relative to center of mass (cm) $\vec{v} = \vec{\omega} \times \vec{r}$, where
 - $\vec{\omega}$: angular velocity
 - $\vec{r}$: displacement vector from cm to point P
- Assume center of mass moves with velocity $\vec{v}_{cm}$, then $\vec{v}_p = \vec{v}_{cm} + \vec{\omega} \times \vec{r}$
- Instantaneous axis has zero velocity, i.e. $v_p = 0$

$$\vec{v}_p = \vec{\omega} \times \vec{r}$$



$$\vec{v}_p = \vec{v}_m + \underbrace{\vec{v}_{p,m}}_{\text{rel velocity}}$$

$$\vec{v}_p = \vec{v}_{cm} + \vec{\omega} \times \vec{r}$$

$$\vec{v}_p = \vec{v}_{cm} - \omega r = 0$$

$$r = \frac{v_{cm}}{\omega}$$

Kinetic Energy of A Rotating Rigid Body

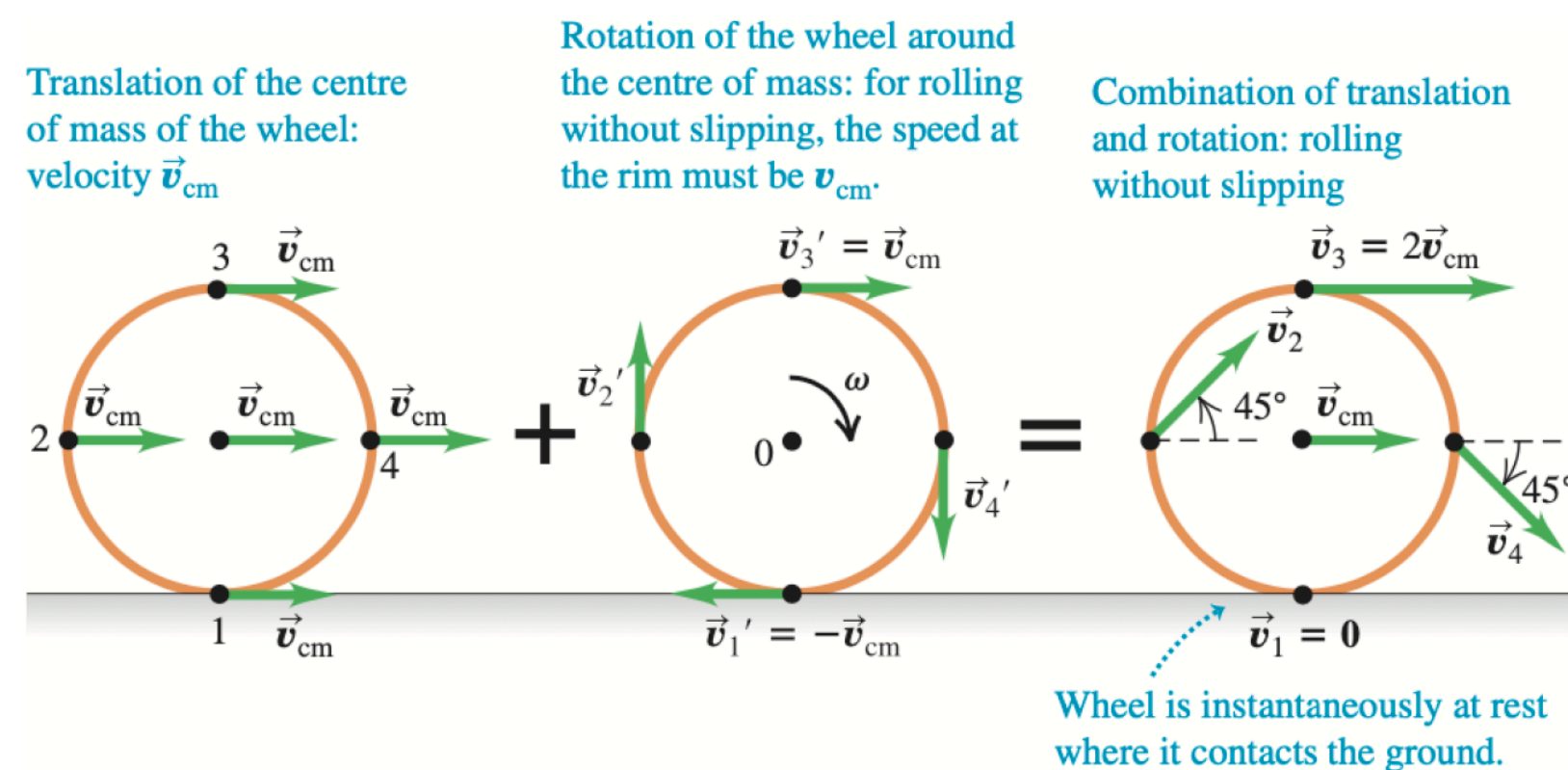
- Use center of mass axis: The total kinetic energy of a rotating rigid body includes translational kinetic energy of center of mass and rotational kinetic energy with respect to center of mass i.e.

$$K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2, \text{ where}$$

- v_{cm} : velocity of center of mass (cm)
 - M : total mass
 - I_{cm} : MOI with respect to (w.r.t) cm
- Use instantaneous axis: the total kinetic energy is purely rotational energy, i.e. $K = \frac{1}{2}I_{\text{insta}}\omega^2$ (HW6, q10)

Roll Without Slipping/Sliding

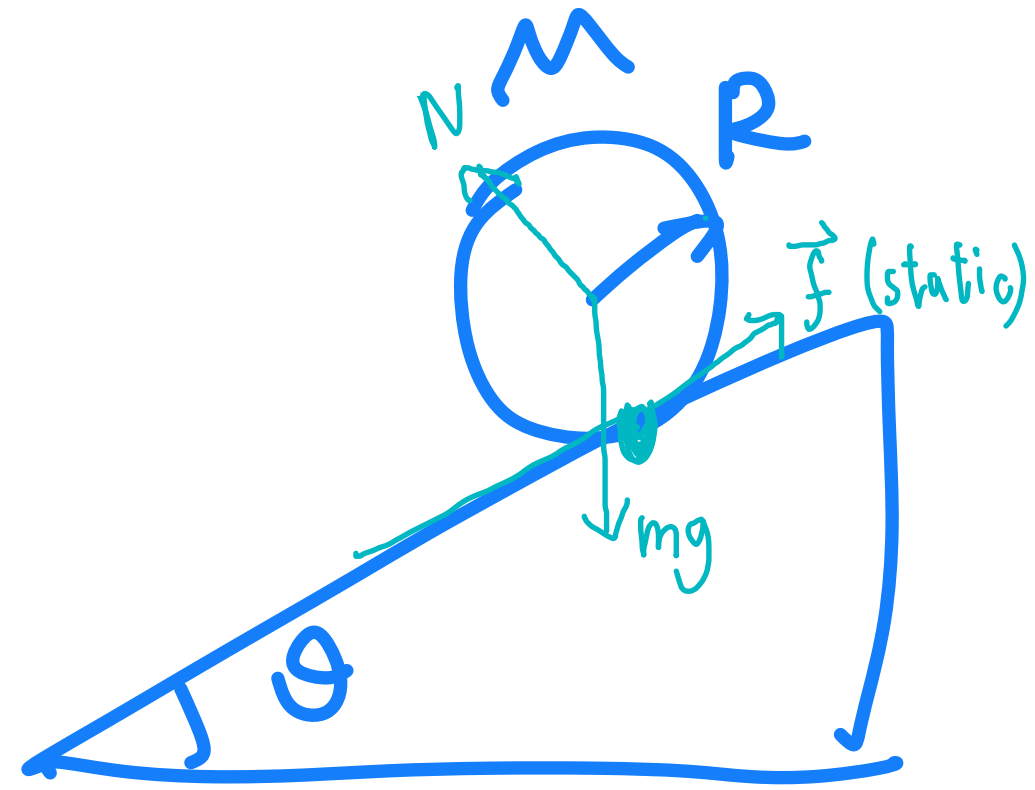
- The condition for rolling with slipping for a circular object (hoop/disk/cylinder/etc) $v_{\text{cm}} = \omega R$, where
 - ω : angular velocity
 - R : radius of the circular object
- This means contact point between object and surface is the instantaneous axis!
- Rolling with slipping also leads to $a_{\text{cm}} = \alpha R$, where
 - a_{cm} : acceleration of cm
 - α : angular acceleration
- Rolling without slipping usually requires sufficient friction from surface.



Acceleration of A Rolling Ball

A solid ball of mass M and radius R rolls down a ramp without slipping. The ramp has an angle θ w.r.t. horizontal plane. The surface is not smooth.

- Find the velocity at the end of the ramp.
- Find the minimum coefficient of static friction μ_s .



$$a) I_{cm} = \frac{2}{5} MR^2$$

$$\text{translation: } Mg \sin \theta - \mu Mg \cos \theta = Ma$$

$$\text{rotation: } \mu Mg \cos \theta \cdot R = \frac{2}{5} MR^2 \cdot \alpha$$

$$\text{roll with slipping: } R\alpha = a$$

$$a = \frac{5}{7} g \sin \theta$$

$$v^2 = 2a \cdot s = 2a \cdot \frac{h}{\sin \theta}$$

$$v_{cm} = \sqrt{\frac{10gh}{7}}$$